Big Data Management
Building a Big Data Architecture

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Introduction and general framework

Introduction - Early 2000s

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- The **Big Data** era had begun, and the profile of the **Big Data engineer** emerged.

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- Open source developers and third parties started looking for ways to **abstract, simplify**, and make big data **available** without the high administrative overhead and cost of managing their clusters.

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Data engineering is increasingly a discipline of **interoperation**; connecting various technologies in streamlined processing workflows to serve ultimate business goals.

Introduction - Project Goal

To build a Big Data Management Architecture

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To build a Big Data Management Architecture

- Understanding the different stages of data management.
- Defining the execution pipeline.
- Selecting the appropriate tools/technologies.
- Deploying and orchestrating the execution.

The Framework

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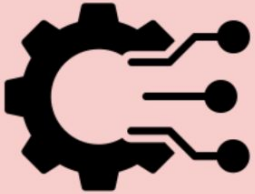
Data Ingestion



Get data from
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The Framework

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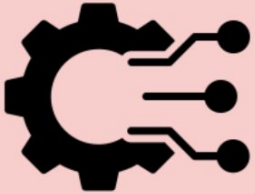
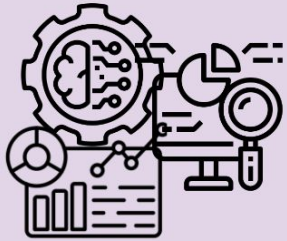
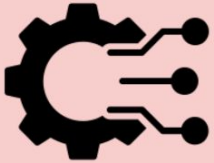
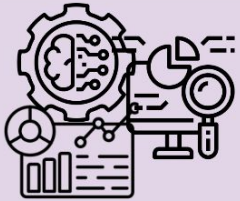
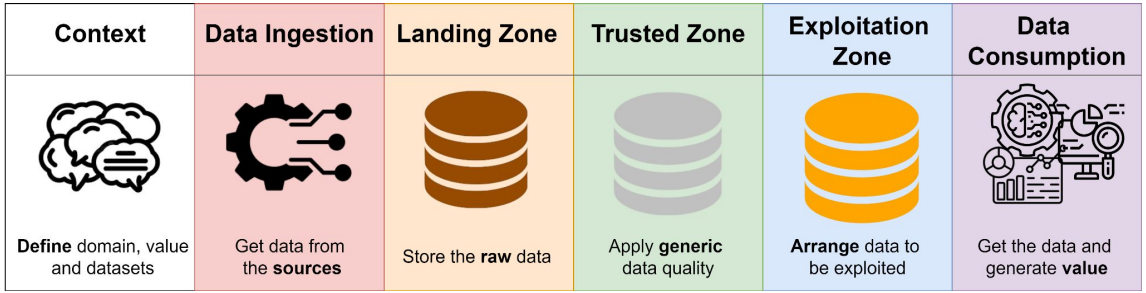
Data Ingestion	Landing Zone	Trusted Zone	Exploitation Zone	Data Consumption
				
Get data from the **sources**	Store the **raw** data	Apply **generic** data quality	**Arrange** data to be exploited	Get the data and generate **value**

The Framework

Context	Data Ingestion	Landing Zone	Trusted Zone	Exploitation Zone	Data Consumption
					
Define domain, value and datasets	Get data from the **sources**	Store the **raw** data	Apply **generic** data quality	**Arrange** data to be exploited	Get the data and generate **value**

The Framework - Considerations

- This framework is meant to **guide** the development of the project, but by any means does it define a hard set of rules to follow.



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- Always make sure to **justify** your decisions.

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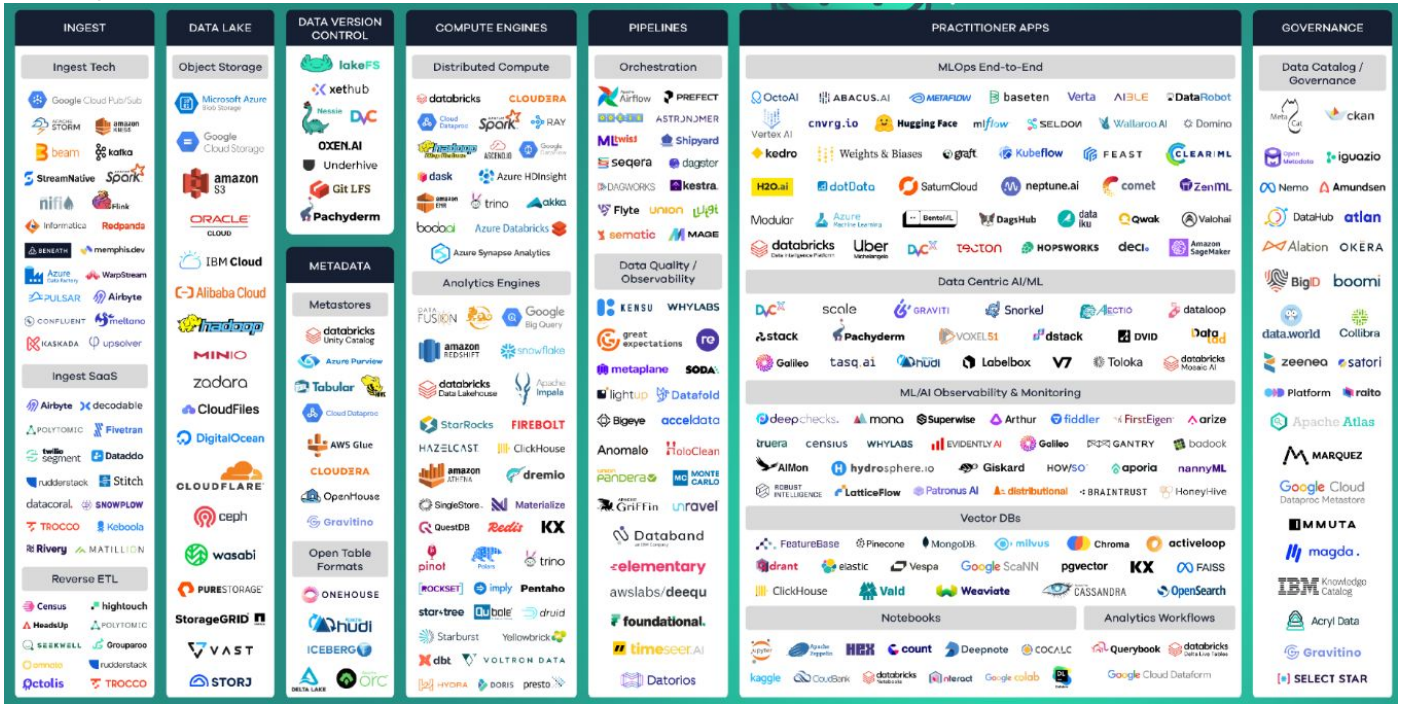
Too many tools



Data management landscape in 2024 ([lakefs])

Organization, deliverables and guidelines

(You have both this and the previous information in LearnSQL)

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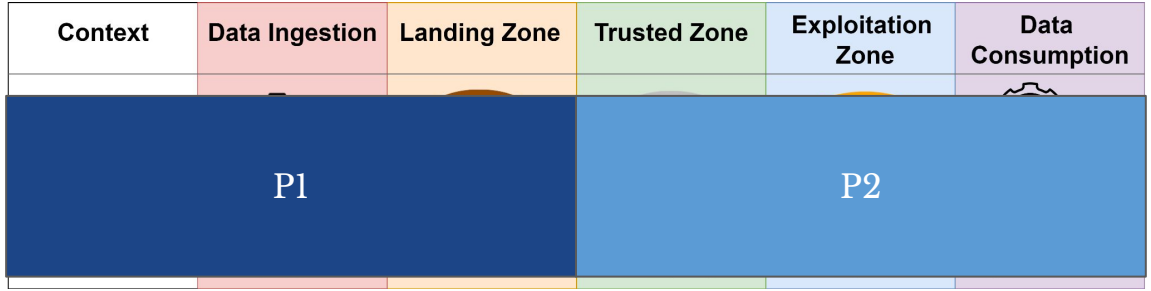

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- In each deliverable we demand **two artifacts**
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- The final grade of the project will be:

$$P = 0.5P1 + 0.5P2$$

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- In each, you will have to include parts of the architecture
- **These will not directly count towards the grade**
  - The goal is to structure the development and provide better feedback
  - If significant effort is show, it will be taken into consideration for the final evaluation

Evaluation guidelines

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- The implementation has to present the desirable properties of software development
  - That is, dynamicity, reusability, openness, reproducibility, etc.

Evaluation guidelines

We want you to be ambitious and develop a complex project